

# PROBLEM-SOLUTION FIT CANVAS 2.0

Project: Script-Controlled ACL - Restrict Record Access Based on Field Value

Domain: ServiceNow System Administration & Security  
Framework: Problem-Solution Fit Methodology (Amaltama)  
Date: October 2026

1. CUSTOMER SEGMENT(S)

CS

- Primary:** ServiceNow System Administrators & Security Engineers.
- Secondary:** IT Service Managers & Compliance Officers.
- Context:** Enterprise organizations handling sensitive record data requiring dynamic, field-level access control.

2. JOBS-TO-BE-DONE / PROBLEMS

J&P

- Job:** Restrict record read/write access dynamically based on runtime field values (e.g., ticket state, assigned group).
- Problem:** OOTB role-based static ACLs cannot handle complex dynamic script conditions across all data interfaces.

3. TRIGGERS

TR

- Security audit findings showing unauthorized record visibility.
- Requirement to lock restricted fields when an Incident/Change enters 'Closed' or 'In Progress' status.

4. EMOTIONS: BEFORE / AFTER

EM

- Before:** Anxious about compliance violations, frustrated by rigid static security roles.
- After:** Confident in platform data governance, assured during security audits.

6. CUSTOMER CONSTRAINTS

CC

- Strict database query performance limits (unoptimized scripts slow down lists/reports).
- Requires security\_admin role for ACL configuration.
- Upgradeability and maintenance constraints.

5. AVAILABLE SOLUTIONS (ALTERNATIVES)

AS

- Client Scripts / UI Policies:** Hide fields visually but fail to secure REST API or Report access.
- Static Role ACLs:** Rigid; creates role explosion for every field state combination.

9. PROBLEM ROOT CAUSE

RC

- Standard declarative ACL Condition Builders lack dynamic evaluation of runtime field states against user properties without server-side script logic.

7. CUSTOMER BEHAVIOUR

BE

- Admins rely on client-side hiding, leaving server-side API security loopholes.
- Manual review of audit logs and security incident tickets when unauthorized access occurs.

8. CHANNELS OF BEHAVIOUR

CH

8.1 Online Channels:

- ServiceNow Developer Portal, Community Forums, Product Documentation, and Admin Workspaces.

8.2 Offline Channels:

- IT Governance & Security Review Boards, Architecture CAB Meetings, Team Sprint Reviews.

10. YOUR SOLUTION: SCRIPT-CONTROLLED ACCESS CONTROL LIST (ACL)

SL

Core Value Proposition & Architecture:

- Server-Side Security Enforcement:** Implements server-side scripted ACL rules (sys\_security\_acl) evaluating runtime field values (e.g., current.state, current.assignment\_group) against session context.
- Universal Access Control:** Enforces data protection across all access layers—Web UI, Next Experience, REST/SOAP APIs, and Reporting Engines.

Implementation & Performance Highlights:

- Optimized Script Logic:** Utilizes lightweight JavaScript evaluation (gs.getUser(), field state checks) without heavy `GlideRecord` database queries inside script blocks.
- Audit-Ready Compliance:** Fully aligned with OWASP access control guidelines and ServiceNow security best practices.

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ServiceNow System Administrator Project Deliverable

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